How Timestral-209M makes a forecast

Mistral AI · 209M parameters · causal transformer with forecast queries · zero-shot

1. One row per series

Targets, past-only and known-ahead covariates. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

Covariate rows only steer the forecast; they are never forecast themselves.

2. Scale as you go, squash, patch

Each row is scaled patch by patch with the mean and standard deviation so far, then squashed by arcsinh. Known-ahead rows use the whole window.

$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x - \mu_{\le k}}{\sigma_{\le k}}\right)$$

value

missing?

one patch = 32 steps

residual MLP

one token: 768 numbers

3. 24 blocks; every 4th reads across rows

After the context tokens come forecast queries: one learned token, repeated once per future patch. RMSNorm in front, shortcut around.

context *forecast queries* (Q Q Q)

Time attention: causal over the context; queries read the context (RoPE, sigmoid gate)

Row attention (6 blocks): down the column; covariates never read the targets

Feed forward: SwiGLU, 768 → 2,048 → 768

4. 1,024 steps in one pass

A head turns each query into **32 steps × 21 quantiles**: the median plus positive steps, so the quantiles never cross.

median

60% band

80% band

The library also forecasts −x and averages the two, and clips at zero a series that was never negative.

Learned forecast queries give the whole horizon in one pass, with quantiles that never cross.

24 blocks (6 read across rows) · width 768 · 12 heads · patches of 32 · context up to 32,768 steps

In the notebooks: last 2,048 days or full monthly history · all covariates and the five metals together · weights: Modified MIT